Retrieval-Augmented Generation (RAG) for Law Firms

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Main Challenges of AI Adoption

Problem with Law Firms

High-Level Confidentiality

Law firms are strictly confidential and have an absolute legal obligation to protect all client data and privacy.

RAG Implementation Barriers

Standard RAG concepts are difficult to directly implement due to strict security and privacy standards foundational to law firms.

External Data Risks

What law firms avoid most is sensitive client documents leaking and being used to train public machine learning AI models.

Information Management & AI

Company’s Knowledge

Knowledge Base & Role of AI

Top-tier law firms increasingly depend on sophisticated knowledge management systems to support their research teams. Today, artificial intelligence empowers these practices to look beyond conventional internal data, facilitating significantly faster and more direct information retrieval.

AI Limitations & Risks

Despite its efficiency, AI presents critical limitations, such as the tendency to "hallucinate" or retrieve synthetically generated information. Furthermore, processing sensitive documents through public AI platforms risks exposing confidential data to external training models, demanding rigorous safeguards to maintain client trust and data security.

Data Security & Flexibility Solution

Closed-Environment RAG

100% Offline & Closed

Runs completely offline within a closed environment. Data and document information will not be shared externally or used as training data.

Model & Chunking Flexibility

Users are free to switch AI models and chunking strategies as needed. No need to rely entirely on a single model.

Independent Local Storage

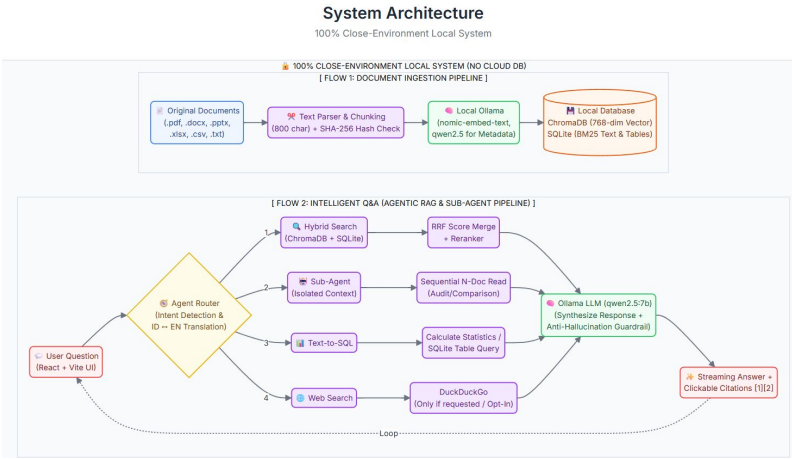
All chunking and AI processing results are stored completely offline within the local system.

System Architecture & How It Works

Closed-Environment Agentic RAG

RAG System Flow Diagram

100% Local Execution (React, FastAPI, Ollama, ChromaDB & SQLite)



Confidentiality Principle: All UI components, server, AI models, and databases run **100% on the local machine** to maintain document security.

Phase 1 — Document Ingestion

- Unique file hash check (**SHA-256**), text extraction, and splitting into small parts (**Chunks**).
- Convert chunks to **768-dimensional** vectors (nomic-embed-text) to store in **ChromaDB** & **SQLite**.

Phase 2 — Intelligent Routing

- **Agent Router** analyzes user queries and selects the best reference:

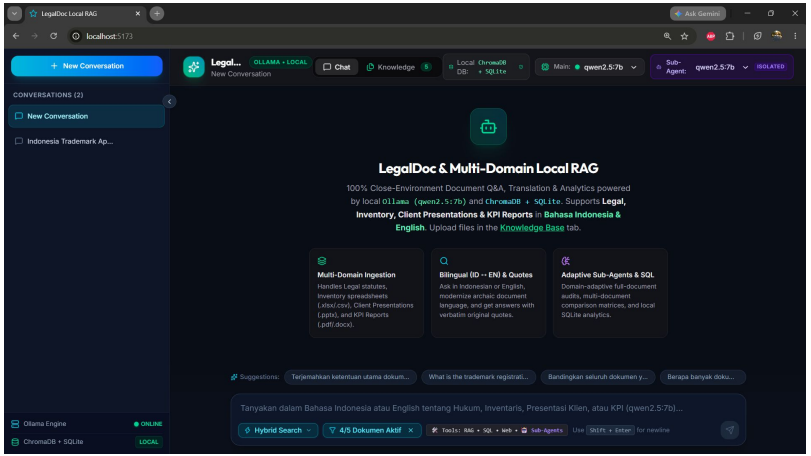
Hybrid Search (articles), **Sub-Agent** (doc audit), **Text-to-SQL** (data analytics), or **Web Search**.

Phase 3 — Grounded Generation & Citation

- Ollama Model (**qwen2.5:7b**) composes the most precise answer in real-time via **SSE Streaming**.
- Features interactive **citation numbers [1] [2]** linked directly to original document text.

User Interface (UI) Design

Clean, Transparent & NotebookLM-Inspired



Designed with a modern dark mode theme (React + Tailwind CSS) that intuitively separates Q&A space and document management.

Dual-View Workspace

Seamlessly switch between Chat and Knowledge Base (a drag-and-drop space for documents).

Model Control & Local Status

Dual AI model selection (Main Agent & Sub-Agent) and real-time connection indicators to Ollama and local databases (ChromaDB + SQLite).

Chat Box Search Control

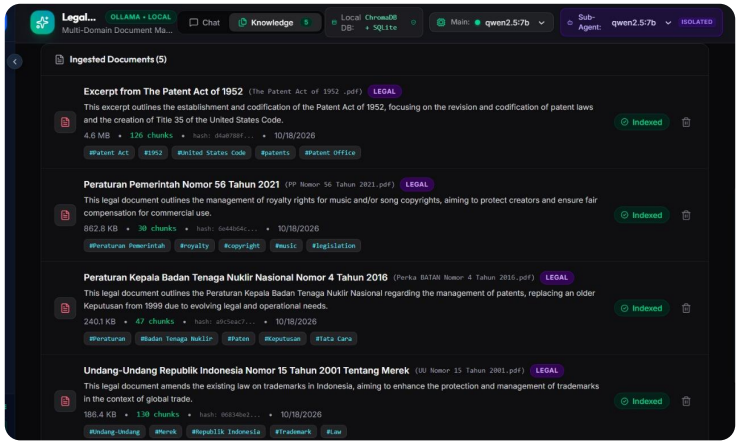
Equipped with prompt suggestions, search strategy options (Hybrid, Vector, Keyword), and Active Document Filter.

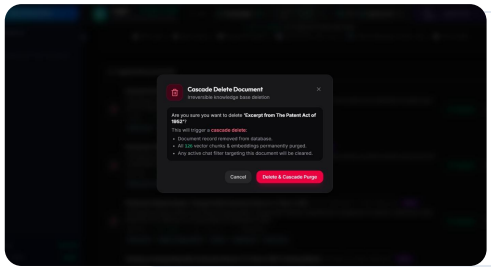
Transparency & Interactive Citations

Displays tool badges, Sub-Agent reasoning process, and clickable citation numbers [1] [2] for verifying original text (preventing hallucinations).

Document Management & Data Processing

Knowledge Base Architecture





1. Smart Deduplication (SHA-256 Hash)

System reads unique file content codes (label hash: d4a0788f...) so identical documents are not processed twice.

2. AI Metadata Extraction

Local **Ollama** model automatically reads document headers to extract **Official Title**, **Summary**, Category (**LEGAL**), and **#keywords**.

3. Chunking & Vector Indexing

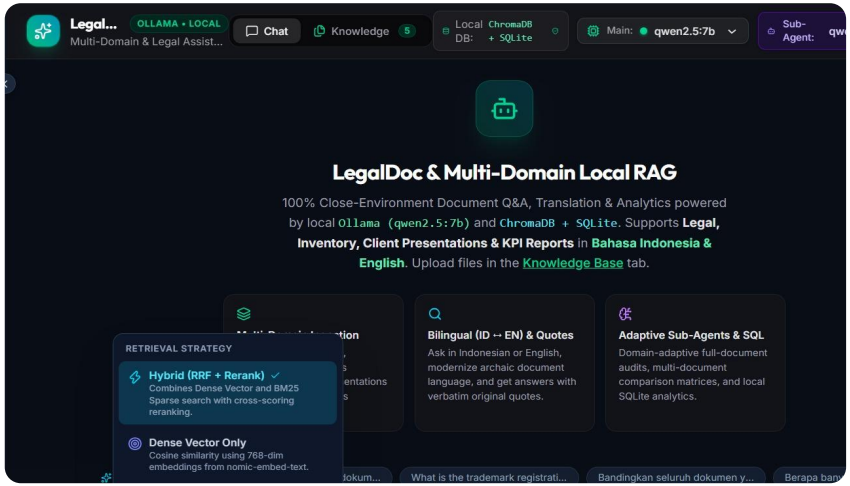
Documents are split into **126 chunks**, converted into vector embeddings (nomic-embed-text), and stored in **ChromaDB & SQLite** (✓ **Indexed**).

4. Data Deletion Security (Cascade Delete)

Equipped with a confirmation *pop-up*. Upon approval, system deletes document record while purging all *chunks* & vectors.

Flexible Search Mode (Retrieval Strategy)

Type of Search & Retrieval Modes



Strategy Selection Interface

The system provides 3 document search methods that can be selected directly according to the user's query context needs.

Hybrid Search (RRF + Rerank) — *Default*

Combines semantic search (*Vector*) and exact keyword search (*BM25*) using *Reciprocal Rank Fusion (RRF)* and *Cross-Scoring Reranker* for maximum accuracy.

Dense Vector Only (ChromaDB)

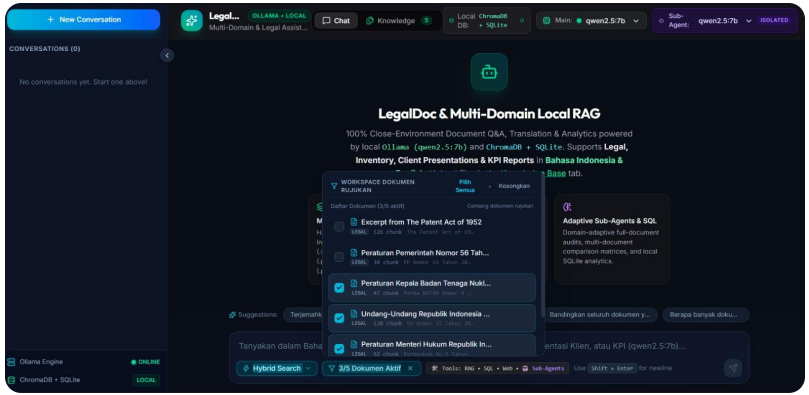
Searches for semantic/contextual similarity using **768-dimensional** vectors (nomic-embed-text). Ideal for conceptual questions and context understanding.

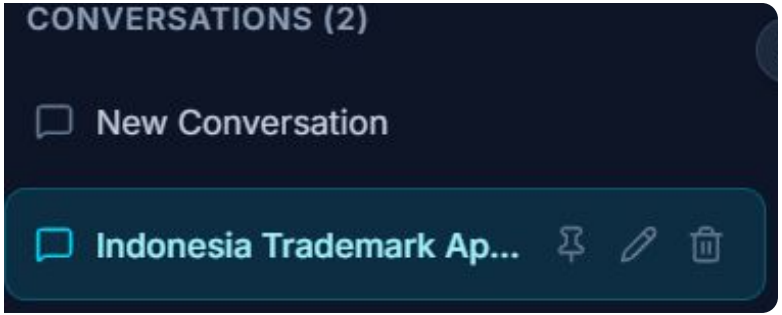
Sparse BM25 Only (SQLite)

Searches for exact keyword matches based on word frequency. Highly effective for exact matching such as article numbers, specific figures, or SKU codes.

Active Document Workspace & Conversation Management

Reference Source Control & Chat Organization





Multi-Document Workspace Filter

Users can select active documents as answer sources (*3/5 Active Documents*) using the **Select All** & **Clear** buttons.

ChromaDB & SQLite automatically restrict searches to selected documents only.

Strict Grounding & Anti-Hallucination

AI is locked to **only** answer based on checked documents.

If information is missing, AI truthfully states data is not found—without making up answers.

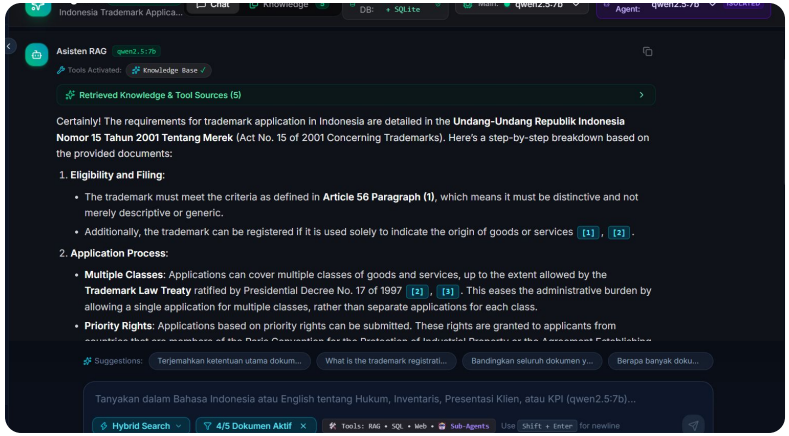
Conversation Management (Pin, Rename, Delete)

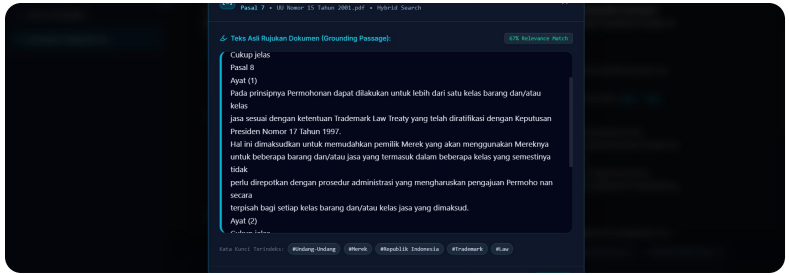
Every discussion session is automatically saved in **SQLite**.

Can be renamed (**Rename**), pinned to the top (**Pin Chat**), or deleted (**Delete**).

Query & Response Interaction (Prompt & Answer)

Prompt Execution Flow & Answer Verification





1. Automatic Contextual Retrieval

When a question is submitted, *Agent Router* activates the **Knowledge Base** ✔ tool and retrieves 7 most relevant chunks from **ChromaDB & SQLite**.

2. Structured Answer & *Streaming*

Local model (**qwen2.5:7b**) composes Indonesian language answers in *real-time* using facts from active documents.

3. Inline Citation Badges ([1], [2])

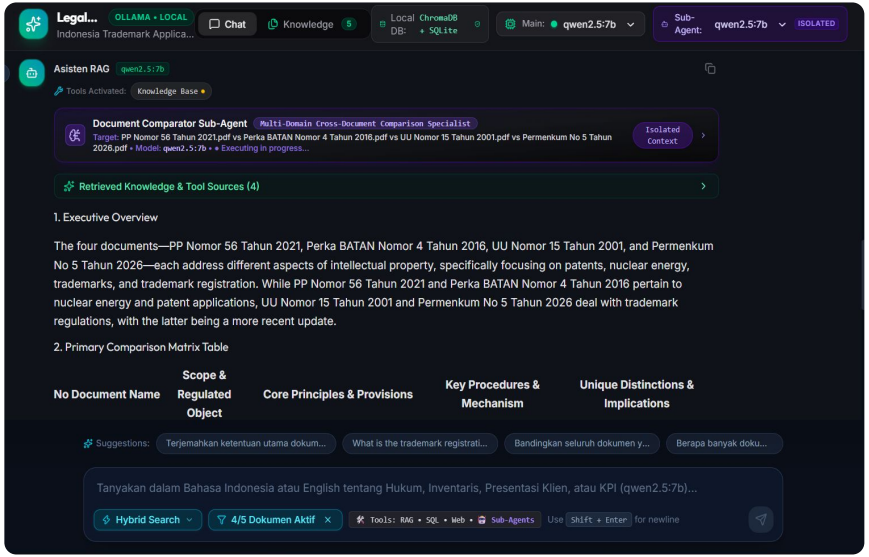
Every claim or legal condition is automatically tagged with interactive, clickable source number indicators.

4. Grounding Passage Inspector

Clicking a badge triggers an original text verification *pop-up* containing Document Title, **Article 2**, **74% Match**, & PDF quotes.

Sub-Agent Features & Comparison Analysis

Automatic Detection & Multi-Document Comparison



1. Automatic Detection & Delegation

Main Agent recognizes comparison instructions and delegates execution to specialist **Document Comparator Sub-Agent**.

2. Dual-Model Selector & Isolated Context

Users can freely switch AI models for *Main Agent* or *Sub-Agent* separately to keep memory overhead optimal.

3. Multi-Document Comparison (NNN-Documents)

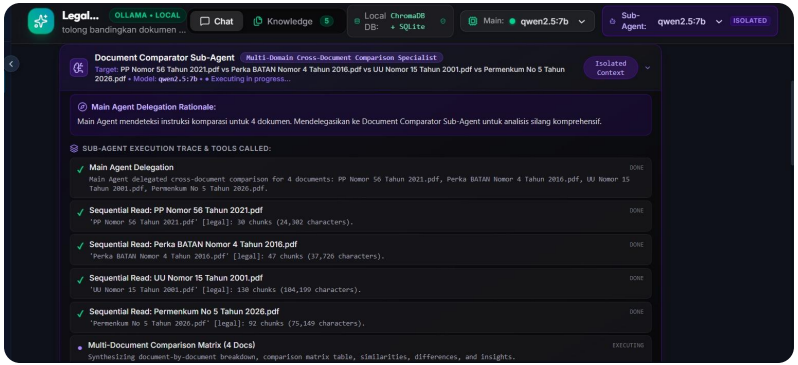
Not limited to 2 files—capable of comparing **multiple documents simultaneously** based on user-activated documents.

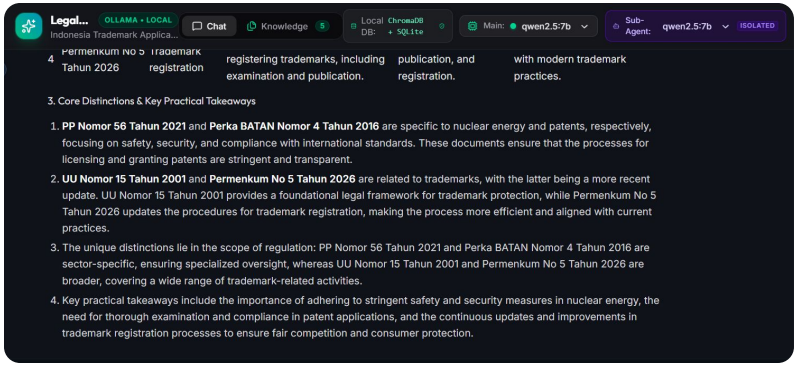
4. Sequential Representative Reading

Sub-Agent reads the initial structure, middle substance, and conclusion of each document (**30, 47, 130, & 92 chunks** across 4 regulations sequentially).

Analysis Results & Comparison Output

Document Comparison - Result





1. Integrated Executive Summary

Outlines the general overview and position of each compared document sequentially.

2. Multilateral Comparison Matrix Table

Presents a side-by-side comparison table (*side-by-side matrix*) so similarities and differences between documents are immediately clear.

3. 4-Dimensional Difference Breakdown

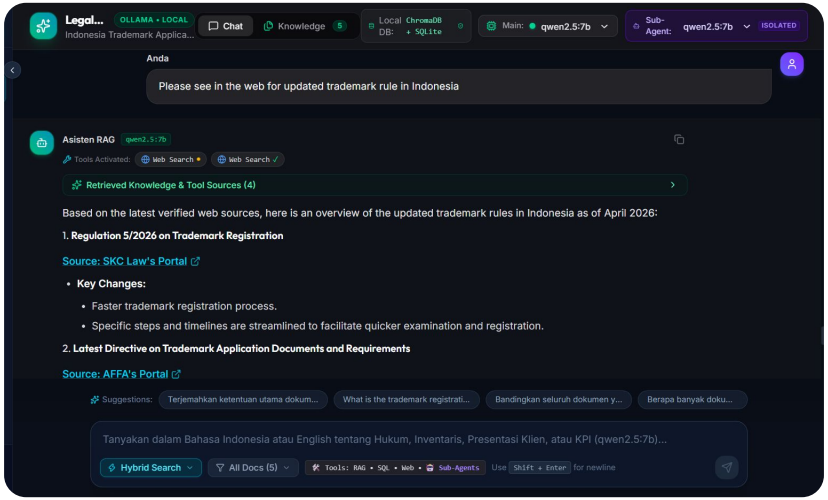
Summarizes analytical conclusions divided into **Main Focus, Regulatory Objectives, Technical Implementation**, and **Practical Effects**.

4. Single-Pass Streaming

The entire report is generated and displayed live (*streaming*) in a single pass, ensuring high speed even on local devices.

Web Search Feature & External Integration

Web Search for External Information



1. Strict Opt-In (Privacy Preserved)

Web Search ✓ tool only runs when users specifically request a web search—local document content remains secure and is never sent outside.

2. Smart Query Extraction

The system cleans conversational command sentences (like *"Please search on the web..."*) and turns them into sharp search keywords.

3. Priority on Official & Trusted Sources

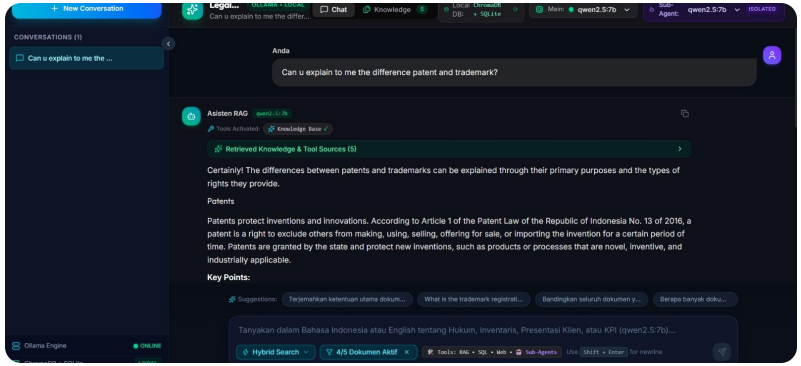
DuckDuckGo search is filtered to prioritize authoritative domains (**.go.id**, BPK, DJKI Ministry of Law) and filter out ad/irrelevant sites.

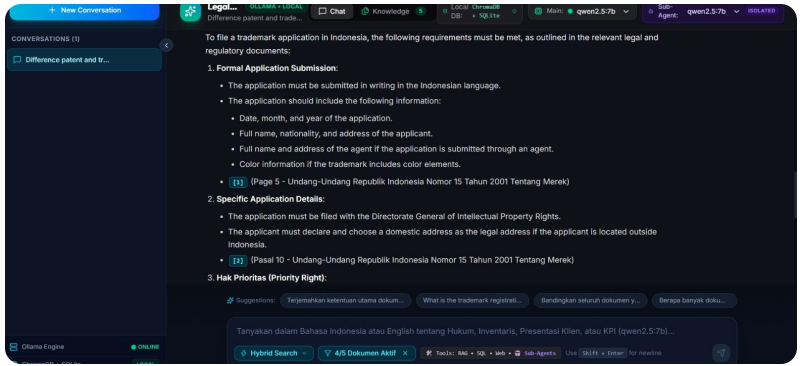
4. Direct Reference Links (Verifiable Links)

Every piece of information from the web must include a **Reference** list with active links (↗) pointing directly to the original source.

Multilingual & Cross-Lingual Features

English & Indonesia Language Capability





1. Automatic Language Mirroring

The system automatically detects the language of the user's *prompt* and adjusts the response language (asking in English → answered in English).

2. Cross-Lingual Term Mapping

Behind the scenes, the *Hybrid Search* engine bridges English–Indonesian legal terminology (e.g., matching *"trademark application"* with *"permohonan pendaftaran merek"*).

3. Contextual & Accurate Translation

AI translates article terms and conditions from Indonesian documents into English (and vice versa) without altering legal meaning.

4. Original Document Citations Preserved

Even if answers are written in English, citation markers **[1]** and **[2]** precisely reference **Page 5 & Article 10 - Law No. 15 of 2001 on Trademarks**.

Summary & Future Outlook

Conclusion & Strategic Roadmap

Key System Achievements

The **LegalDoc Local RAG System** successfully delivers an enterprise-grade, privacy-first legal AI platform. By combining local LLM capabilities with advanced sub-agent workflows, it transforms complex legal document analysis into an efficient, highly reliable process.

1. End-to-End Privacy & Security: 100% local execution using Qwen 2.5 models ensures sensitive legal documents remain strictly on-premises.

2. Sub-Agent Multi-Doc Comparison: Automated delegation reads and synthesizes complex multi-regulation comparisons in a single pass.

3. Cross-Lingual & Grounded RAG: Seamlessly bridges English and Indonesian legal domain terms with exact citation tracking.

4. Controlled External Integration: Privacy-preserved opt-in web search focused strictly on trusted government domains (.go.id).

Future Expansion Roadmap

To further advance AI-driven legal operations, the platform is strategically positioned to integrate specialized local enhancements, expand document intelligence, and strengthen corporate governance controls.

1. Domain Fine-Tuning: Fine-tuning open-source LLMs specifically on the Indonesian legal corpus (UU, Perpres, & court rulings) for deeper statutory comprehension.

2. Advanced Legal OCR Engine: Deep integration of high-precision OCR pipeline to automatically parse scanned court decisions and legacy archive PDFs.

3. Role-Based Access Control (RBAC): Division-level permissions ensuring strict data segregation across corporate legal, compliance, and HR teams.

4. Continuous Evaluation Loop: Automated benchmark suite to measure RAG retrieval recall, accuracy, and hallucination prevention continuously.